



ARTIFICIAL INTELLIGENCE

Complete Learning Roadmap

Build intelligent systems with modern AI concepts, model workflows, and responsible deployment.

Python foundations

Supervised learning

Unsupervised learning

Model evaluation

Feature engineering

PyTorch/TensorFlow

01

Fundamentals

02

Core Concepts

03

Tools and Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Artificial



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Python foundations

02

Core Concepts

Supervised learning •
Unsupervised learning • Model
evaluation • Feature
engineering

03

Tools and Frameworks

PyTorch/TensorFlow

04

Real Projects

Classifier project • RAG
assistant project • Model API
service • Evaluation
dashboard

05

Advanced Topics

Responsible AI • Model
governance • Latency
optimization • Production drift
monitoring

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Python foundations
- Linear algebra essentials
- Statistics fundamentals
- Data preprocessing

Tools & Frameworks

- Python foundations

Recommended Projects

- Artificial Intelligence workshop using Python foundations
- Artificial Intelligence portfolio project with Linear algebra essentials

Stage 1 Milestone Complete Artificial Intelligence workshop using Python foundations and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Supervised learning
- Unsupervised learning
- Model evaluation
- Feature engineering

Tools & Frameworks

- Supervised learning
- Unsupervised learning
- Model evaluation
- Feature engineering

Recommended Projects

- Artificial Intelligence workshop using Supervised learning
- Artificial Intelligence portfolio project with Unsupervised learning

Stage 2 Milestone Complete Artificial Intelligence workshop using Supervised learning and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- PyTorch/TensorFlow
- Experiment tracking
- Prompt engineering
- Vector databases

Tools & Frameworks

- PyTorch/TensorFlow

Recommended Projects

- Artificial Intelligence workshop using PyTorch/TensorFlow
- Artificial Intelligence portfolio project with Experiment tracking

Stage 3 Milestone Complete Artificial Intelligence workshop using PyTorch/TensorFlow and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Classifier project
- RAG assistant project
- Model API service
- Evaluation dashboard

Tools & Frameworks

- Classifier project
- RAG assistant project
- Model API service
- Evaluation dashboard

Recommended Projects

- Artificial Intelligence workshop using Classifier project
- Artificial Intelligence portfolio project with RAG assistant project

Stage 4 Milestone Complete Artificial Intelligence workshop using Classifier project and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

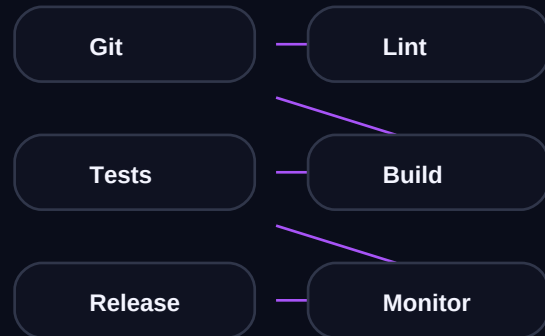
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Responsible AI
- Model governance
- Latency optimization
- Production drift monitoring

CI/CD Pipeline



Stage 5 Milestone

A production-grade artificial intelligence workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Artificial Intelligence docs and community guides

Link: <https://platform.openai.com/docs>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/ai-data-scientist>

PAID

Structured specialization track focused on Artificial Intelligence

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Artificial Intelligence roadmap reference

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Official documentation

Link: <https://platform.openai.com/docs>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Artificial Intelligence?
- Which project best proves Stage 4 competency in Artificial Intelligence?
- What trade-offs appear in Stage 5 decisions for Artificial Intelligence?